DAA-SPEC-0.2.0

DESKTOP AI AGENT

Technical Specification · Demo Release

DOCUMENT CONTROL

DOCUMENT ID	DAA-SPEC-0.2.0
REVISION	0.2.0
CLASSIFICATION	INTERNAL / DEMO
PLATFORM	Windows 10 / 11 · Python · 3.10 · Ollama
ENTRY POINT	python -m app | scripts\start.bat
PACKAGE	src/desktop_ai_agent
RUNTIME DEPS	Installed automatically on every start (no requirements.txt)

SYSTEM PIPELINE



SCOPE: Local multimodal desktop assistant. Gemma 4 (Ollama) plans tool calls; gated action layer executes allowlisted operations; OverlayBar + optional TTS output.

01 OVERVIEW

Local multimodal desktop assistant. User intent (text or voice) enters the orchestrator; Gemma 4 via Ollama emits tool calls; the action layer validates and executes only registry-approved tools; results return as chat text and optional speech.



Feedback loop: ACTION results · ORCHESTRATOR (self-correction until stop)

CAPABILITY MATRIX · DEMO V0.2.0

DOMAIN	STATUS	IMPLEMENTATION
Chat UI	IN DEMO	Always-on-top OverlayBar; context meter; Stop / New chat / Config
Voice in	IN DEMO	Push-to-talk + wake word via faster-whisper
Core AI	IN DEMO	Ollama gemma4 / gemma4-agent; tool calling; folder index
Desktop actions	IN DEMO	open_app, web_*, screenshot, wait, UI click/type
Voice out	IN DEMO	edge-tts or offline SAPI
Safety gate	IN DEMO	Registry + executor; HIGH risk denied (no confirm UI)
Startup deps	IN DEMO	bootstrap_deps pip-install on every launch
Browser automation	OUT	No Playwright / form filling
Camera / gestures	OUT	Not implemented
Code execution	OUT	provide_script is display-only

02 ARCHITECTURE

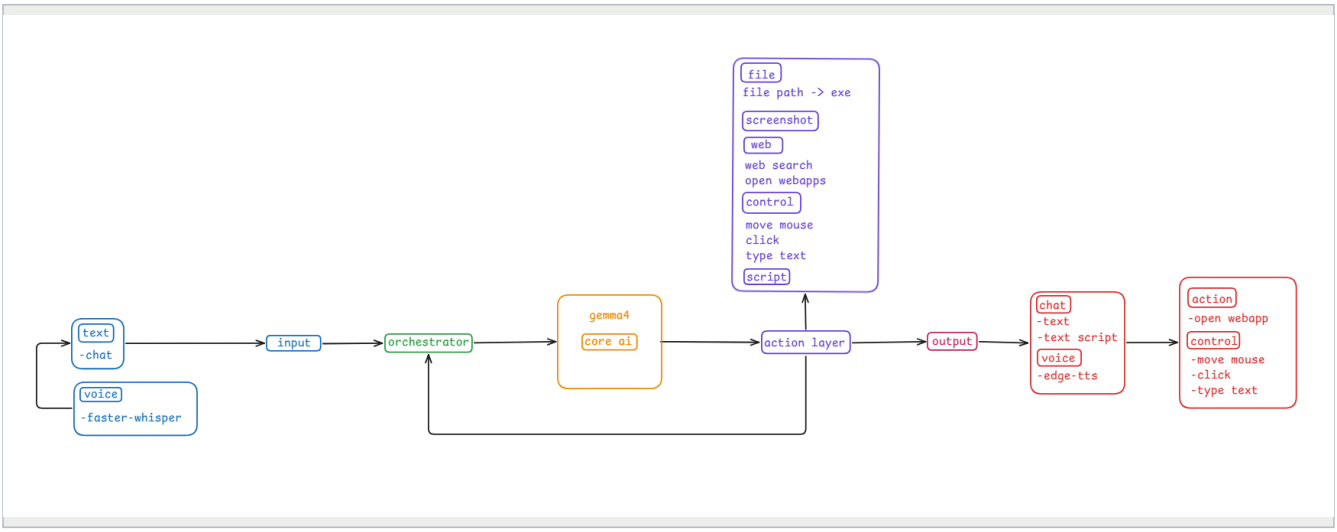
LAYER RESPONSIBILITIES

LAYER	ROLE	KEY MODULES
Bootstrap	Install runtime deps every start	bootstrap_deps.py
Input	Text + STT / wake · task text	input/voice.py, ui_elements.py
Orchestrator	Agent loop; never touches OS	agent_loop, service, prompts, task_state
Core AI	Ollama chat + app catalog	session_setup, chat_client, folder_index/
Action	Validate · risk · run · audit	registry, executor, handlers, tools/
Output	Optional speech	output/tts.py
UI	Setup window + OverlayBar	launcher, overlay_bar, chat_text

PACKAGE MAP · SRC/DESKTOP_AI_AGENT/

- bootstrap_deps.py · pip install runtime packages on every start
- main.py · DPI awareness + LauncherApp
- input/ · voice STT, UI Automation listing
- core/ · Ollama, session, monitors, folder_index, chat_client
- orchestrator/ · agent_loop, AgentService, prompts, task_state
- action/ · registry, executor, handlers, audit_log, tools/
- output/tts.py · edge / sapi speaker
- ui/ · launcher, overlay_bar, icons + SVG assets

REFERENCE DIAGRAM



03 QUICK START

PREREQUISITES

ITEM	REQUIREMENT
OS	Windows 10 / 11
Runtime	Python 3.10+
LLM host	Ollama installed and on PATH

LAUNCH SEQUENCE

```
python -m app
# or:  scripts\start.bat  |  scripts\start.ps1
```

Every start: (0) pip-install runtime deps (1) detect monitors / DPI (2) index apps on C/D/E (3) ensure Ollama + models (4) show OverlayBar.

SMOKE PROMPTS

PROMPT	EXPECTED PATH
open steam / open vs code	open_app via executable index
what's the weather in Budapest?	web_search · answer from snippets
open youtube and play lofi	web_open https URL
wait 2 seconds	wait
what do you see on my screen?	auto screenshot + vision
click the Start button	list_elements · click
create a file on my desktop	provide_script (display only)

04 CORE AI

Core does not touch the OS. SessionSetup.prepare(): index apps · ensure Ollama · pull base model · create gemma4-agent from Modelfile · apply /set commands · ChatClient.

OLLAMA CONFIGURATION (SETUP.JSON · OLLAMA)

KEY	DEFAULT	PURPOSE
base_model	gemma4	Pulled if missing
agent_model	gemma4-agent	Built from config/Modelfile.gemma4-agent
host	http://127.0.0.1:11434	Ollama HTTP API
startup_timeout_sec	60	Wait for daemon ready

NOTE	Live system instructions live in orchestrator/prompts.py (edit without rebuilding the model). Chat options: temperature 0.2, num_ctx 8192.

FOLDER INDEX

Sources: Start Menu .lnk, Uninstall registry, AppX (AUMID), Program Files/(x86), Games folders, extra_roots. Output: config/program_files_index.json (+ .txt). open_app launches catalog targets only · the model never invents exe paths.

05 ORCHESTRATOR

AgentService is the UI façade: submit / stop / reset on a background thread. Callbacks must be marshalled to the Tk thread.

HARD LIMITS

PARAMETER	VALUE	EFFECT
max_rounds	20	Stop with status max_rounds
max_failed_rounds	2	Consecutive step failures · failed
max plan steps	10	Executor rejects oversized plans
CONTEXT_TOKENS	8192	UI meter + num_ctx
HANDOFF_AT	~85%	Compact / restart chat
MAX_HANDOFFS	3	Cap on context handoffs

CONTROL BEHAVIOURS

- Vision heuristic · auto-capture when user message looks like screen Q&A
- Deferred screenshot · ·open X and see·· skips initial capture; refresh after UI change
- Follow-ups · after click / type_text, auto list_elements(active)
- Nudge · one corrective prompt if model promises an action without a tool call
- Leaked call recovery · parse Gemma call:name{·} text into real tool_calls
- Element list cache · mouse/type actions use latest list_elements (cleared after click/type)

06 ACTION LAYER

Only the action layer may change the machine. Flow: model tool calls · plan_from_tool_calls · Executor · handlers · results · model.

REGISTERED ACTIONS

ACTION	RISK	SPECIFICATION
screenshot	LOW	primary|all|monitor|region; agent windows excluded from capture
wait	LOW	seconds 0·30
web_search	LOW	ddgs snippets; max_results 1·5
list_elements	LOW	area active|taskbar|desktop|all; ~60 items / ~60s TTL
provide_script	LOW*	Display-only JSON; ends_turn; never executed
open_app	MED	Indexed launchers only; no launch args
web_open	MED	http/https URL only
move_mouse	MED	By element number · model never sends raw coords
click	MED	left|right; optional double; rate-limited
type_text	MED	Paste into inputs; no Enter; no passwords

SAFETY INTERLOCKS

- Unknown actions / bad args · rejected; reason returned for model retry
- Whole plan validated before any side effect
- Risk levels set in registry by maintainers · model cannot change them
- HIGH risk requires confirm; confirm UI not wired · denied by default
- Registry names must match HANDLERS or executor refuses to start
- UI control: click gap 1s / 10 per min; corner emergency stop 30s; risky-name regex
- Screenshots use SetWindowDisplayAffinity (exclude agent UI)
- Audit trail: logs/audit.jsonl · rejected | declined | execute | result

EXTEND THE REGISTRY

```
1. Add ActionSpec in action/registry.py
2. Add handler + HANDLERS entry in action/handlers.py
3. Add tests under tests/action/
```

07 USER INTERFACE

COMPONENT	FUNCTION
LauncherApp	Setup log window; wires voice / TTS / AgentService / OverlayBar
OverlayBar	Always-on-top top-edge bar: collapsed · expanded · reply
Controls	Input, Send, Mic, Stop, New chat, Settings, Close
Context meter	8k window; warn ·75%; full ·90%
Auto-hide	ui.bar_auto_hide_ms (default 20000)

08 VOICE & TTS

VOICE (SETUP.JSON · VOICE.*)

Keys: enabled, model_size, language, sample_rate, device, compute_type, wake_word.{phrases, stop_phrases, silence_sec, energy_threshold, ·}. Wake phrase · wake_ack TTS · task session · stop/sleep/goodbye.

TTS (SETUP.JSON · OUTPUT.TTS.*)

KEY	NOTES
engine	edge (neural, needs internet) | sapi (offline Windows)
voice / rate / volume	Engine-specific voice id and speed
wake_ack	Spoken after wake (default: yes my lord?)
enabled	Master switch

09 CONFIGURATION

Primary file: config/setup.json. Legacy key program_files_index still accepted.

SECTION	CONTROLS
ollama	models, host, timeout, interactive_commands
folder_index	drives, apps.* include flags, max_exe_depth, max_exes
ui	setup size, bar_width, bar_auto_hide_ms, bar_collapsed_height, bar_top_margin
voice	STT + wake listening
output.tts	speaker engine and voice
formats	Legacy {script,code} notes · prefer tool calls
screen.monitor	Optional; used by SessionSetup monitor preference

HARDCODED — Agent-loop limits, CONTEXT_TOKENS, executor max_steps, UI-control rate rules, and audit path are code defaults · not setup.json keys.

10 STARTUP DEPENDENCIES

No requirements.txt. bootstrap_deps.ensure_dependencies() runs on every launch and pip-installs runtime packages (keep in sync with pyproject.toml extras).

```
Pillow  pymupdf  ddgs
faster-whisper  sounddevice  numpy
mss  comtypes  pyttsx3  edge-tts  pygame
```

Dev only (manual): pip install -e ".[dev]" then pytest

11 TESTING

```
pip install -e ".[dev]"
pytest
```

Suites: tests/{core,action,orchestrator,ui,input,output}. Stdlib + pytest. Voice / UI Automation unit tests mock optional deps; folder index uses tmp_path fakes.

12 OPERATIONS & LIMITS

ARTIFACT	LOCATION / NOTE
Audit log	logs/audit.jsonl
Screenshots	logs/screenshots/*.png
App index	config/program_files_index.json
Model risk	Small local model may pick wrong tool · check audit
Wake word	Rolling Whisper, not dedicated wake engine
type_text	Never submits (no Enter); skips password fields

13 REPOSITORY LAYOUT

```
desktop_ai_agent/
··· app/                      # python -m app path bootstrap
··· src/desktop_ai_agent/     # package
··· config/                   # setup.json, Modelfile, indexes
··· docs/Demo_v0.2.0.pdf      # this manual
··· scripts/                  # start.* + generate_docs_pdf.py
··· tests/
··· pyproject.toml
```

14 STATUS

REV	Demo release 0.2.0. Expect rough edges with a small local model. Single PDF supersedes the previous multi-file docs/ set.